6 The applicants qualified for position y_i is detailed:

position	qualified
y_1	x_1, x_4, x_8, x_9
y_2	$x_1\ x_8\ x_7$
y_3	$x_3\ x_6\ x_7\ x_8$
y_4	x_3
y_5	x_4, x_{10}
y_6	$x_1\ x_2\ x_5$
y_7	$x_2\ x_5$

An SDR is indicated filling all positions. This can't be beat.

7 The family doesn't have an SDR, because the subfamily $\mathcal{A}' = \{A_1, A_3, A_4, A_5\}$ has

$$\left|\bigcup_{A_i \in \mathcal{A}'} A_i\right| = 3 < 4 = |A_i|$$

so fails the marriage condition. So by Thm 9.2.2 there is no SDR.

8 There are two. More generally, the family

$$\mathcal{A} = (A_1, \ldots, A_n) \text{ where } A_i = \{i, i+1\}$$

for $i = 1 \ldots n-1$, has two SDRs. Indeed, if A_1 is represented by 1, then A_2 must be represented by 2, and for all i A_i must be represented by i. Alternately, if A_1 is represented by 2, A_i must be represented by $i+1 \pmod n$ for all i.

No.

9 Let $\mathcal{A} = (A_1, \dots, A_n)$ be a family of sets with an SDR $E = (e_1, \dots, e_n)$ and x be a member of A_1. If $x \in \{e_1, \dots e_n\}$ then E is an SDR containing x. Otherwise, $(x, e_2, e_3, \dots, e_n)$ is an SDR containing x

The family $\mathcal{A} = (A_1, A_2)$ with $A_1 = \{x, y\}$, $A_2 = \{x\}$ has the SDR (y, x) but no SDR in which x represents A_1.

17 Assume there is a marriage in which A is paired with $b \neq a$, and a is paired with $B \neq A$. Then $A \leftrightarrow b$, $B \leftrightarrow a$, $A >_a B$, and $a >_A b$. So the marriage is unstable by definition. So no such marriage is stable.